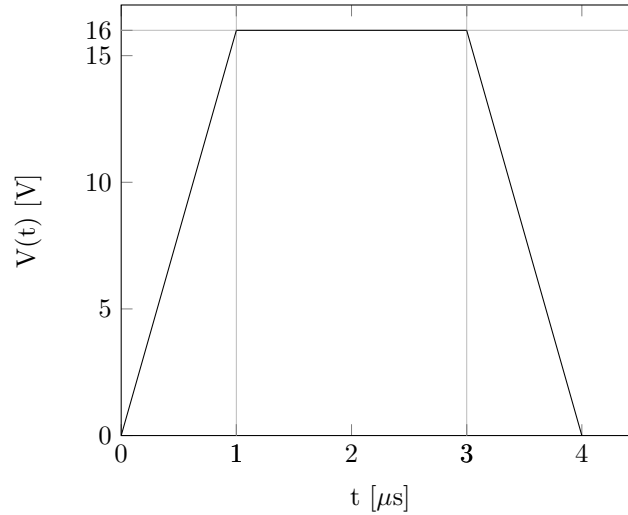


Tutorial 7: Memo

Question 6.10.



01	The voltage through a 5 mF capacitor is shown. Determine I_C between $t = 0$ and $4\mu\text{s}$. [2]
-----------	---

Memo

$$I_C = C \frac{dV}{dt}$$

For $0 \mu\text{s} \leq t \leq 1 \mu\text{s}$:

$$\begin{aligned} I_C &= (5 \times 10^{-3}) \times \frac{d}{dt} \left(\frac{16t}{1 \times 10^{-6}} \right) \\ &= (5 \times 10^{-3}) (16 \times 10^6) \\ &= 80 \text{ kA} \end{aligned}$$

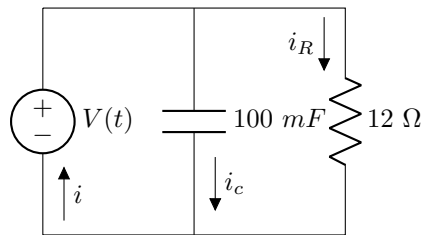
For $1 \mu\text{s} \leq t \leq 3 \mu\text{s}$:

$$\begin{aligned} I_C &= (5 \times 10^{-3}) \times \frac{d}{dt} (16) \\ &= (5 \times 10^{-3}) (0) \\ &= 0 \text{ A} \end{aligned}$$

For $3 \mu\text{s} \leq t \leq 4 \mu\text{s}$:

$$\begin{aligned} I_C &= (5 \times 10^{-3}) \times \frac{d}{dt} \left(64 - \frac{16t}{1 \times 10^{-6}} \right) \\ &= (5 \times 10^{-3}) (-16 \times 10^6) \\ &= -80 \text{ kA} \end{aligned}$$

Question 6.12.



02 If $V(t) = 45e^{-2000t}$ V, find the power of the voltage source. [2]

Memo

Find the current through the capacitor (i_C):

$$\begin{aligned} i_c(t) &= C \frac{dV}{dt} \\ &= (100 \times 10^{-3}) (-2000 \times 45e^{-2000t}) \\ &= -9000e^{-2000t} \text{ A} \end{aligned}$$

Find the current through the resistor (i_R):

$$\begin{aligned} i_R(t) &= \frac{V(t)}{R} \\ &= \frac{45e^{-2000t}}{12} \\ &= 3.75e^{-2000t} \text{ A} \end{aligned}$$

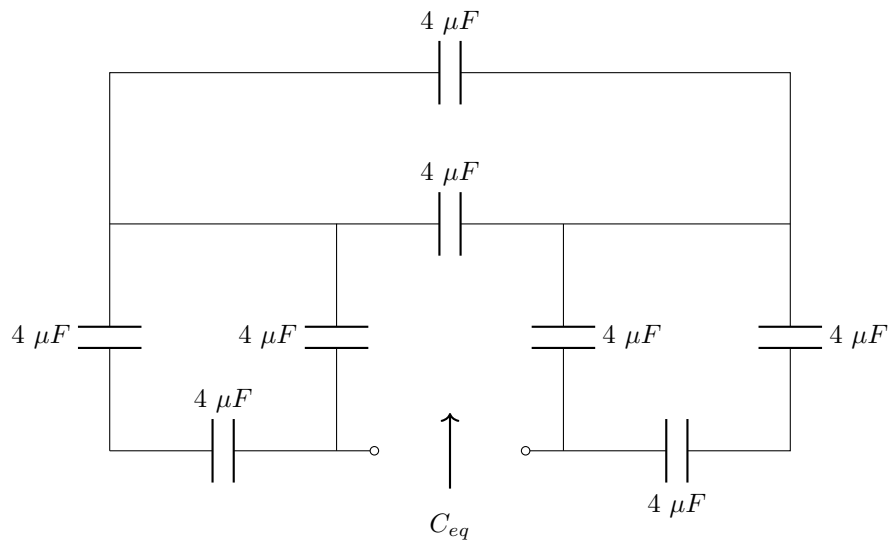
Find the total current (i):

$$\begin{aligned} i(t) &= i_c(t) + i_R(t) \\ &= -9000e^{-2000t} + 3.75e^{-2000t} \\ &= -8996.25e^{-2000t} \text{ A} \end{aligned}$$

Finally, find the power:

$$\begin{aligned} p(t) &= V(t)i(t) \\ &= (45e^{-2000t}) (-8996.25e^{-2000t}) \\ &= -404.83e^{-4000t} \text{ kW} \end{aligned}$$

Question 6.18.



03 Find C_{eq} [2]

Memo

Series capacitors equation:

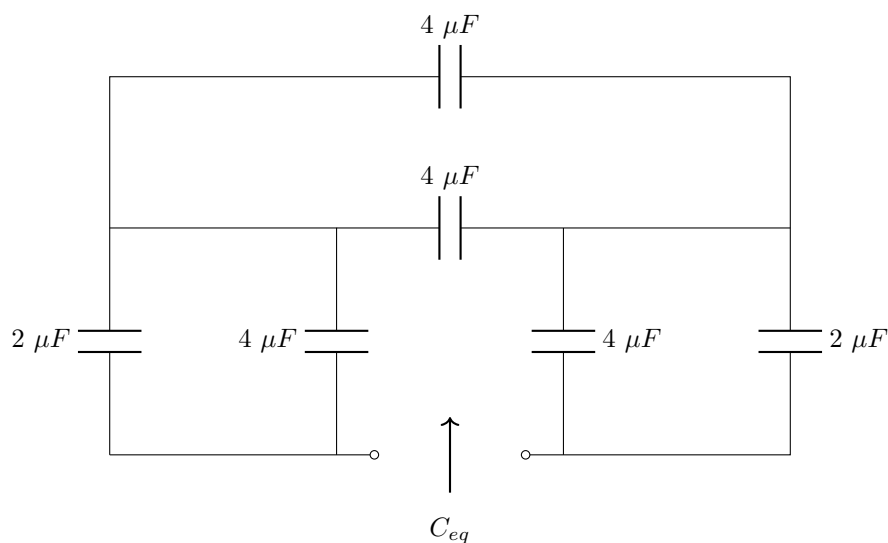
$$C = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n} \right)^{-1}$$

Parallel capacitors equation:

$$C = C_1 + C_2 + C_3 + \dots + C_n$$

Combine the series capacitors on the bottom right and bottom left:

$$\left(\frac{1}{4 \times 10^{-6}} + \frac{1}{4 \times 10^{-6}} \right)^{-1} = 2 \mu F$$

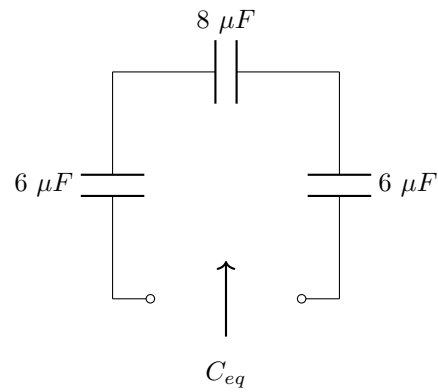


Combine the parallel capacitors at the top:

$$4 \mu F + 4 \mu F = 8 \mu F$$

Combine the parallel capacitors on the bottom left and right:

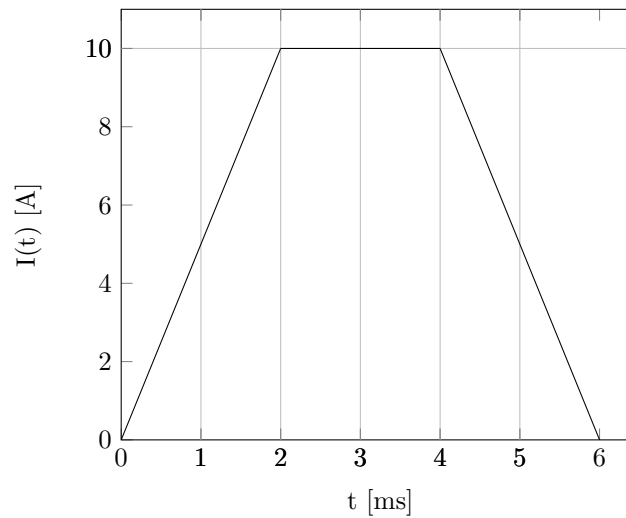
$$2 \mu F + 4 \mu F = 6 \mu F$$



Combine remaining series capacitors:

$$\begin{aligned} C_{eq} &= \left(\frac{1}{6 \times 10^{-6}} + \frac{1}{8 \times 10^{-6}} + \frac{1}{6 \times 10^{-6}} \right)^{-1} \\ &= 2.1818 \mu F \end{aligned}$$

Question 6.40.



04 The current through a 5 mH inductor is shown. Determine V_L at $t = 1, 3$ and 5 ms . [2]

Memo

The equation for voltage across an inductor:

$$V_L = L \frac{dI}{dt}$$

At 1 ms:

$$\begin{aligned} V(1\text{ms}) &= (5 \times 10^{-3}) \times \frac{d}{dt} \left(\frac{(10 - 0)t}{2 \times 10^{-3} - 0} \right) \\ &= (5 \times 10^{-3}) (5 \times 10^3) \\ &= 25\text{ V} \end{aligned}$$

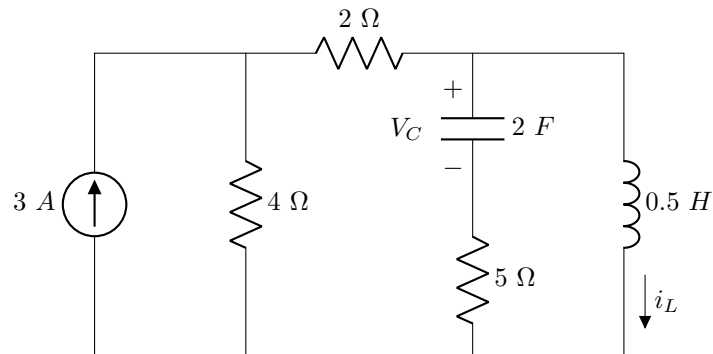
At 3 ms:

$$\begin{aligned} V(3\text{ms}) &= (5 \times 10^{-3}) \times \frac{d}{dt} (10) \\ &= (5 \times 10^{-3}) (0) \\ &= 0\text{ V} \end{aligned}$$

At 5 ms:

$$\begin{aligned} V(5\text{ms}) &= (5 \times 10^{-3}) \times \frac{d}{dt} \left(30 - \frac{(0 - 10)t}{6 \times 10^{-3} - 4 \times 10^{-3}} \right) \\ &= (5 \times 10^{-3}) (-5 \times 10^3) \\ &= -25\text{ V} \end{aligned}$$

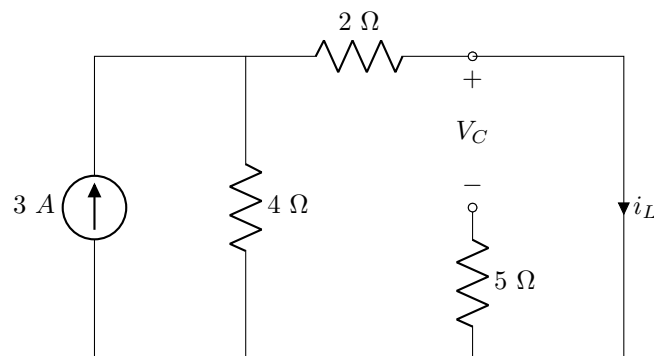
Question 6.46.



05 Determine V_C , i_L , W_C and W_L using DC conditions [4]

Memo

Under DC conditions capacitors become open circuits and inductors become short circuits.



Due to the inductor being a short circuit:

$$V_C = 0 \text{ V}$$

Calculate i_L using current division:

$$\begin{aligned} i_L = i_{2\Omega} &= 3 \left(\frac{4}{2+4} \right) \\ &= 2 \text{ A} \end{aligned}$$

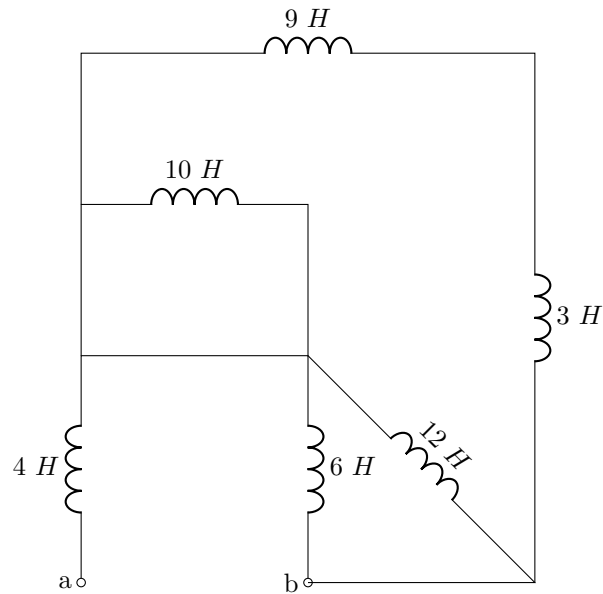
Calculate W_C :

$$\begin{aligned} W_C &= \frac{1}{2} C V_C^2 \\ &= \frac{1}{2} (2) (0)^2 \\ &= 0 \text{ J} \end{aligned}$$

Calculate W_L :

$$\begin{aligned} W_L &= \frac{1}{2} L i_L^2 \\ &= \frac{1}{2} (0.5) (2)^2 \\ &= 1 \text{ J} \end{aligned}$$

Question 6.54.



06 Find L_{eq} across a - b [2]

Memo

Equation for series inductors:

$$L = L_1 + L_2 + L_3 + \dots + L_n$$

Equation for parallel inductors:

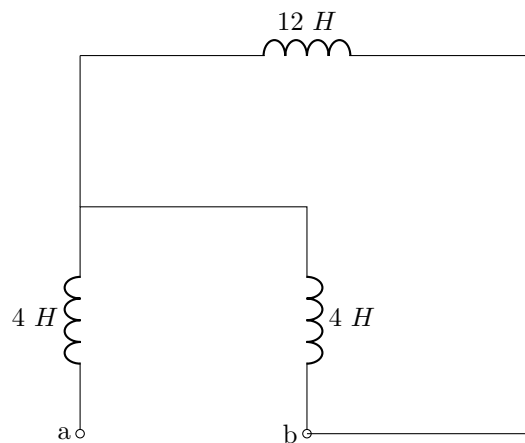
$$L = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n} \right)^{-1}$$

The 10 H inductor is shorted. The 9 H and 3 H inductors are in series:

$$9 \text{ H} + 3 \text{ H} = 12 \text{ H}$$

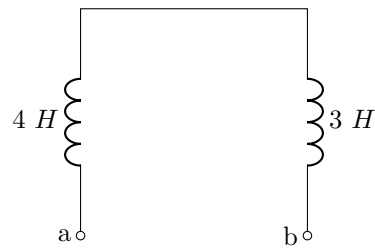
The 12 H and 6 H inductors are in parallel:

$$\left(\frac{1}{12} + \frac{1}{6} \right)^{-1} = 4 \text{ H}$$



The 12 H and 4 H inductors are in parallel:

$$\left(\frac{1}{12} + \frac{1}{4} \right)^{-1} = 3 \text{ H}$$



Finally, the 4 H and 3 H inductors are in series:

$$L_{eq} = 4 \text{ H} + 3 \text{ H} = 7 \text{ H}$$